Class 18 van der Pol oscillator

Activity

Teams 3 & 4, post photos of your work to the class Google Drive (see Canvas for link). Make a folder for today's class if one doesn't exist yet.

1. (convince yourself of the details: van der Pol example) After a change of variables the van der Pol system is

$$\dot{x} = \mu(y - \frac{1}{3}x^3 + x)$$
$$\dot{y} = -\frac{1}{\mu}x.$$

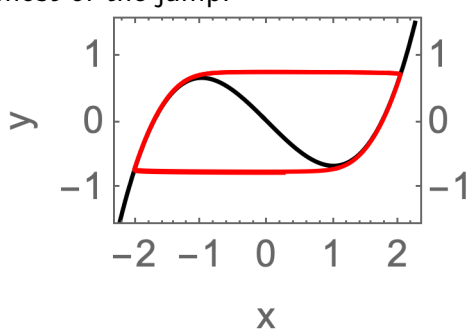
Consider the case where $\mu \gg 1$.

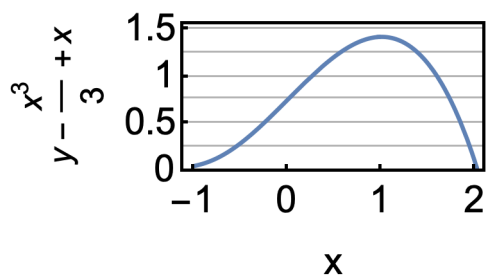
(a) In this relaxation oscillation, the trajectory is moving quickly when it jumps between the two parts of the $\dot{x} = 0$ nullcline.

In the plot on the left, the red curve is the trajectory.

In the plot on the right, the blue curve is showing $y - \frac{1}{3}x^3 + x$ as a function of x along the upper jump, where $y \approx 2/3$, $1 \leq x \leq 2$.

By looking at the values of $y - \frac{1}{3}x^3 + x$ in the graph, convince yourself that $\dot{x} > 0.1\mu$ for most of the jump.





We say this velocity is of the order of μ, or is $\mathcal{O}(\mu)$, because it is bounded above by a constant multiple of μ.

(b) The trajectory is traversing a distance of about 3 as it jumps. Use the nullcline to find the start and end point of the jump to confirm the distance.

$\langle \dot{x} \rangle \approx \mu/3$ along the jump.

Combine the approximate distance and the approximate average velocity to find the μ-dependence of the time that a particle spends jumping.

(c) While moving along the nullcline, the trajectory moves about 1 unit in x and a bit less than 2 units in y. It is basically moving along the curve $y = \frac{1}{3}x^3 - x$ (it is not quite on the curve, but it is close to that curve the whole time). The time it spends traversing the curve is a constant multiple of μ^k for some integer k.

To estimate time (just as we did for the oscillators in chapter 4), we set up an integral of the form $\displaystyle\int_{x_1}^{x_2} \frac{dt}{dx} dx$ or something like this. This integral shows us how the time depends

on μ. Using

$$\int_{x_1}^{x_2} \frac{dt}{dx} dx$$

doesn't work so well. It puts $y - (\frac{1}{3}x^3 - x)$ in the denominator (so a dependence on x and y, not just x). What is wrong with having a dependence on x and on y in the integral?

(d) We could try again with

$$\int_{y_1}^{y_2} \frac{dt}{dy} dy.$$

Argue that this leads to a problem, too, and isn't something we can integrate.

(e) So actually, Steve used

$$\int_{x_1}^{x_2} \frac{dt}{dy}\frac{dy}{dx} dx.$$

(This is an example of persisting until something works, and luckily getting something to work before we run out of options). Confirm that this expression results in something that is integrable.

Note that for $\frac{dy}{dx}$ we're thinking of trajectories as, to good approximation, being stuck on the nullcline, so compute this by assuming the trajectory is exactly on the nullcline.

(f) Use the setup of this final integral to identify k. There is no need to evaluate the integral. We just want to learn the μ dependence of the time.

(g) Compare the amount of time spent jumping to the amount of time spend moving along the curve. (We have the timescales of these processes, and not the exact amounts of time, so compare the timescales).

2. Set up an integral in a single variable for the amount of time it takes a trajectory to traverse the curve $y = F(x)$ from $x = x_0$ to $x = x_1$, when $\dot{y}$ is given by $\dot{y} = g(x) = x$ and $F(x) = x^2 - 2x$

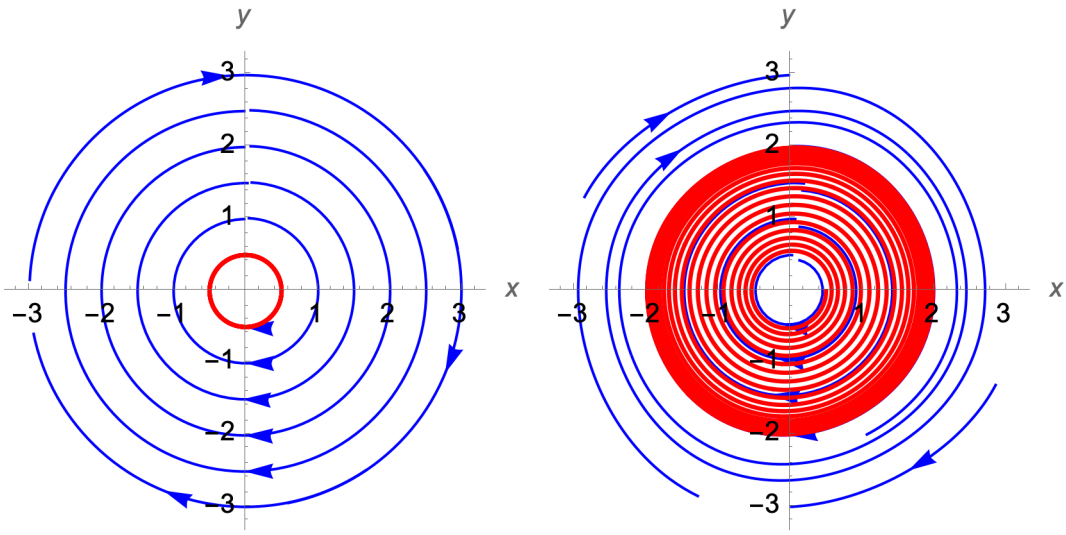
3. (weakly nonlinear van der Pol)

Let $E(x, \dot{x}) = \frac{1}{2}x^2 + \frac{1}{2}\dot{x}^2$ or $E(x, y) = \frac{1}{2}x^2 + \frac{1}{2}y^2$

(a) Find $\frac{dE}{dt}$ for the weakly nonlinear van der Pol oscillator, where $x + \ddot{x} = -\epsilon\dot{x}(x^2 - 1)$.

Write $\dot{E}$ in terms of just x and $\dot{x}$ or y.

(b) Along a trajectory, we have $\Delta E = \int_0^T \frac{dE}{dt} dt$. The weakly nonlinear van der Pol is very close to the system $\ddot{x} + x = 0$. Two trajectories are shown for each system in the plots below.



In the $\ddot{x} + x = 0$ system, the period of cycles is $T = 2\pi$, and $x(t) = A\cos t$ is a solution for any A.

In the weakly nonlinear van der Pol system, assume there exists a limit cycle (we know that there is one from Lienard's theorem), and that it is of the form $x(t) = A\cos t$ with period $T \approx 2\pi$. Find the value of A associated with the limit cycle via the following steps:

- Assume $x(t) = A\cos t$ on the limit cycle, and use this $x(t)$ to find $\dot{x}$.
- Substitute $\dot{x}$ and x into
  $\Delta E \approx -\epsilon \int_0^{2\pi} \dot{x}^2(x^2 - 1)dt$
- Using $\frac{1}{2\pi}\int_0^{2\pi} \sin^2 t \; dt = \frac{1}{2}$ and $\frac{1}{2\pi}\int_0^{2\pi} \cos^2 t \sin^2 t \; dt = \frac{1}{8}$, find a value of A such that $\Delta E = 0$.
- Check the A you found against what is happening in the phase portrait above.

4. (Ruling out closed orbits) Let

$$\begin{aligned} \dot{x} &= y \\ \dot{y} &= x - x^3 - \mu y, \quad \mu \geq 0. \end{aligned}$$

This system is known as the unforced Duffing oscillator. (Duffing published work on this kind of equation in 1918, and it became a popular model in the 1970s). The μy term is a damping term.

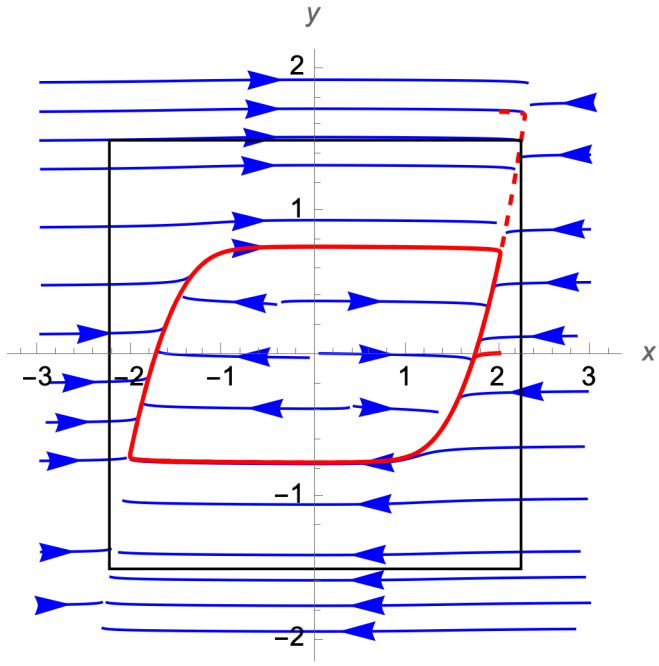
(a) This system can also be written as $\ddot{x} = x - x^3 - muy$. How does this differ from the similar form that we have seen for a conservative system?

(b) Use Bendixson's criterion to show that the system has no closed orbits for $\mu > 0$.

5. Find a restriction on a and b such that $V(x, y) = ax^2 + by^2$ is a Liapunov function for

$$\begin{aligned} \dot{x} &= y - x^3 \\ \dot{y} &= -x - y^3 \end{aligned}$$

6. (trapping region for van der Pol) Consider the rectangular box shown in the figure below. The upper right and lower left corners co-incide with the cubic nullcline.



Sketch the nullclines and a vector representing the direction of the vector field in each of the four sectors created by the nullclines. Why isn't the box shown above a region that traps all trajectories? Where are trajectories able to escape the region?